

# Parental Job Loss and Children's Health:

Ten Years after the Massive Layoff of the SOEs'  
Workers in China

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# Summary

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## □ Research Question:

- Does forced job loss cause poor children's health?

## □ Main Findings:

- Paternal job loss has significant negative effect on children's health.
  - Maternal job loss has **no** significant effect.
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# I. Introduction

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- ☐ **Children's health is an important kind of human capital.**
  - ☐ Significantly related to educational performance and long-term labor market outcomes in later life.
- ☐ **However, the theory of parental employment and children's health is ambiguous:**
  - ☐ Parental employment brings more financial resources
  - ☐ Employment likely reduces the parental time with the children.

# I. Introduction

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- ❑ China began to reform its Soviet-style economic system in 1978.
  - ❑ In the **mid 1990's**, China initiated a profound reform of state-owned enterprises (SOEs), and started to **massively lay off redundant state workers** (Appleton, Song, and Xia 2005).
  - ❑ From 1998 to 2005, more than **35 million** workers got laid off
    - ❑ The total number of workers in the state sector went down from 144 million to 78 million (World Bank, 2007).
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# II. Theory and Channels

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## Children's Health as Economic Capital

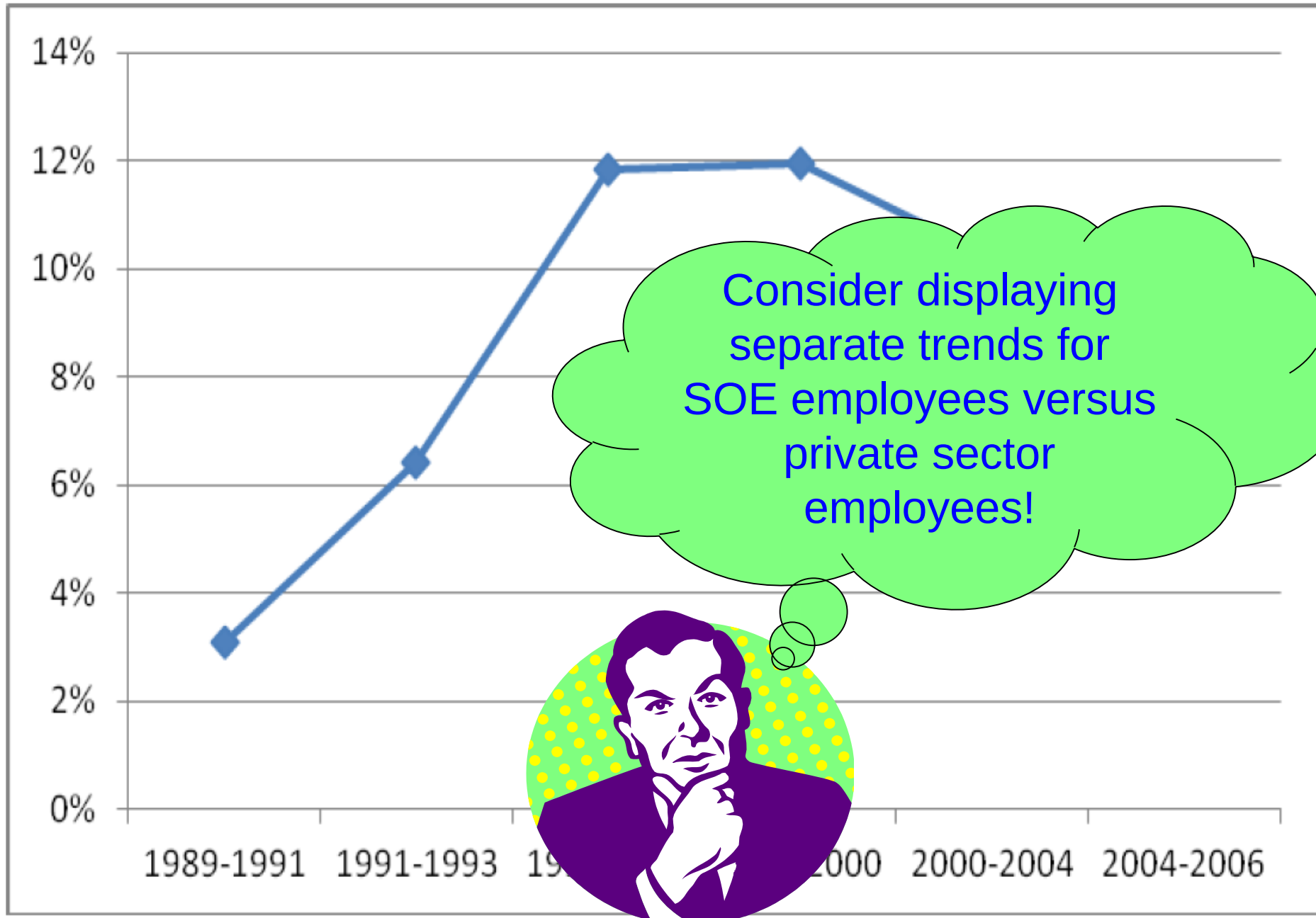
- ❑ Health is a form of human capital. But unlike other forms of capital health cannot be purchased.
  - ❑ Conceptually, health can be modeled as an output of household production (*inputs*).
  - ❑ In this model, households “combine time and market goods to produce more basic commodities that directly enter their utility functions.” (Becker 1965).
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## II. Theory and Channels

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- ❑ There is an ambiguous effect of parental job loss on child health status,
    - ❑ **Income loss**: Job loss might result in lost income and decrease other inputs to child health (e.g. nutrition).
    - ❑ **Time allocation**: Displaced parents can spend more time at home to provide better supervision and care.
    - ❑ **Lifestyle changes**: Unemployed parents may suffer from stress and so engage in risky behaviors.
  - ❑ **Heterogeneity**: Maternal job loss may differ from the effect of paternal job loss.
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\ **Figure 1. The Trend of Self-Reported Job Loss (CHNS 1989-2006)**



# III. Data

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- ❑ The China Health and Nutrition Survey (CHNS) is a panel data of 19,000 individuals over 1989 to 2009.
  
- ❑ **Measures of Children's Health**
  - ✓ Height-for-age z-score (HAZ), and
  - ✓ Weight-for-age z-scores (WAZ)

# IV. Framework

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**Sources of endogeneity:** Evaluating the causal effect of job loss on childhood health outcomes.

- ❑ **Non random assignment**: Those who have lower work productivity may have higher chances of being laid off and are also more likely to have children with worse health status.
- ❑ **Simultaneity**: The probability of job loss may be effected by the child's health status.
- ❑ **Measurement error**: job loss cannot be measured directly and must be inferred from the data.

# IV. Framework

## Identification Strategy:

- ❑ The state run enterprise (SOE) retrenchment is an exogenous shock for individual households.
- ❑ Fixed effects control for unobserved time-invariant household background.
- ❑ Proxy variables for parent's productivity are included as controls.
- ❑ Instrumental variables are identified for *jobloss*.

**Identifying Assumption:** (a) unobserved characteristics related to job loss are time-invariant, and (b) instruments are valid and exogenous.

# IV. Framework

- The main econometric model is a fixed-effects panel data model.

$$H_{iht} = \beta_0 + \beta_1 \text{jobloss}_{ht} + \beta_2 H_{ijt-1} + \beta_3 X_{iht-1} + \beta_4 W_{h,t-1} + \beta_5 v_t + \mu_i + \epsilon_{iht}$$

- The lagged dependent variable is added to control for the child's health status before their parent lost the job.

# IV. Framework

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- ❑ “The strict exogeneity assumption never holds in unobserved effects models with lagged dependent variables”  
(Wooldridge 2006, p. 290)
- ❑ Solution: Dynamic GMM (e.g. Arellano & Bond 1991 REStud). See Wooldridge 2006, section 11.6.2.
- ❑ References: Nickell (1981) or Matyas & Balazsi (2012)

# IV. Framework

□ The first stage equation is specified as:

$$\begin{aligned}
 jobloss_{iht} &= \alpha_0 + \alpha_1 Z_{1h,t-1} + \alpha_2 Z_{2c,t-1} + \alpha_3 H_{ih,t-1} \\
 &+ \alpha_4 X_{ih,t-1} + \alpha_5 W_{h,t-1} + \alpha_6 v_t + \eta_i + e_{iht}
 \end{aligned}$$

- **SOE employment:** binary variables indicating whether both parents worked in SOEs.
- **Local job opportunities:** (i) The county-level layoff rate; (ii) indicator for open trade near the neighborhood; (iii) number of enterprises in the community.

# IV. Framework

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- ❑ 2SLS is biased towards OLS when (a) the instruments are weak and (b) when there are many overidentifying restrictions.
- ❑ The **limited information maximum likelihood** (LIML) estimator is approximately median-unbiased for overidentified constant effects models (Angrist and Pischke 2009)
- ❑ LIML has the advantage of having the same asymptotic distribution as 2SLS (under constant effects) – but is a less precise estimator.

# V. Empirical Results



|                                 | Fixed Effect + IV<br>(LIML, robust to weak instruments) |                           |
|---------------------------------|---------------------------------------------------------|---------------------------|
|                                 | Height-for-age<br>z-score                               | Weight-for-age<br>z-score |
|                                 | (1)                                                     | (2)                       |
| <b>Panel A: Father</b>          |                                                         |                           |
| Paternal job loss               | -0.68*<br>(0.39)                                        | -0.22<br>(0.43)           |
| First-stage <i>F</i> -statistic | $F=8.28$<br>$P<0.001$                                   | $F=8.31$<br>$P<0.001$     |
| Overidentification test         | $P=0.540$                                               | $P=0.699$                 |
| Endogeneity test                | $P=0.288$                                               | $P=0.856$                 |
| No. of observations             | 1075                                                    | 1075                      |
| <b>Panel: Mother</b>            |                                                         |                           |
| Maternal job loss               | -1.15<br>(0.95)                                         | -1.12<br>(0.88)           |
| Weak instrument test            | $F=2.26$<br>$P=0.049$                                   | $F=2.28$<br>$P=0.046$     |
| Overidentification test         | $P=0.359$                                               | $P=0.976$                 |
| Endogeneity test                | $P=0.306$                                               | $P=0.245$                 |
| No. of observations             | 1075                                                    | 1075                      |

Consider interacting the  
excluded instruments  
with the included  
variables!



# V. Empirical Results



|                   | Height-for-age<br>z-score |          | Weight-for-age<br>z-score |          |
|-------------------|---------------------------|----------|---------------------------|----------|
| Paternal Job Loss | -0.33**                   | -0.34**  | -0.38**                   | -0.34**  |
| HAZ(t-1)          |                           | -0.17*** |                           |          |
| WAZ(t-1)          |                           |          |                           | -0.32*** |
| Maternal Job Loss | -0.28                     | -0.26    | -0.17                     | -0.21    |
| HAZ(t-1)          |                           | -0.14*** |                           |          |
| WAZ(t-1)          |                           |          |                           | -0.31*** |

**Interpretation:** Paternal job loss is associated with a 1/3 standard deviation\* decrease in height for age, and a 1/3 standard deviation\* decrease in weight for age.

There is no mention if  
standard errors were  
corrected (e.g. cluster on  
region)



# V. Empirical Results

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- ❑ Placebo test “as though” job loss occurred one year prior to actual job loss.
  - Not statistically significant.

# V. Empirical Results

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- ❑ **Dependent variable:** Household income
  - Paternal job loss decreases income but maternal job loss does not.
  
- ❑ **Dependent variable:** Fat, Protein, Calories
  - Paternal job loss decreases nutrition but maternal job loss does not.

# V. Empirical Results

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- ❑ **Dependent variable:** Immunizations, preventive care, at-home childcare, home cooked meals
  - Maternal job loss increases time spent on care but paternal job loss does not.
  
- ❑ **Dependent variable:** Smoking
  - Paternal job loss increases the probability of risky behaviors but maternal job loss does not.

# VI. Conclusions

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- ❑ Fail to reject exogeneity of job loss using weak instruments
  - ❑ All the results are presented as though job loss is exogenous.
- ❑ Conflict between fixed effects (FE) and lagged dependent variable (LDV) research design.
  - ❑ We recommend considering Dynamic GMM.
- ❑ Cluster standard errors?



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# VI. Channels

|        | <b>Chng.<br/>Household<br/>Income<br/>(1000s Yuan)</b> | <b>Chng.<br/>Household<br/>Income</b> | <b>Chng. Per<br/>Capita<br/>Household<br/>Income</b> | <b>Chng. Per<br/>Capita<br/>Household<br/>Income</b> |
|--------|--------------------------------------------------------|---------------------------------------|------------------------------------------------------|------------------------------------------------------|
| Father | -8.34***                                               |                                       | -2.34***                                             |                                                      |
| Mother |                                                        | -1.34                                 |                                                      | -0.30                                                |

# VI. Channels



|                                            | Protein           |                          | Fat               |                          | Calorie           |                           | Carbohydrate      |                        |
|--------------------------------------------|-------------------|--------------------------|-------------------|--------------------------|-------------------|---------------------------|-------------------|------------------------|
|                                            | (1)               | (2)                      | (3)               | (4)                      | (5)               | (6)                       | (7)               | (8)                    |
| <b>Panel A: Father</b>                     |                   |                          |                   |                          |                   |                           |                   |                        |
| Paternal job loss                          | 0.12<br>(0.09)    | 0.25*<br>(0.14)          | -0.00<br>(0.09)   | 0.09<br>(0.14)           | 0.09<br>(0.10)    | 0.22<br>(0.14)            | 0.07<br>(0.09)    | 0.09<br>(0.14)         |
| Paternal job loss<br>×low income( $t-1$ )  |                   | <b>-0.60**</b><br>(0.24) |                   | <b>-0.49**</b><br>(0.24) |                   | <b>-0.64***</b><br>(0.24) |                   | <b>-0.33</b><br>(0.23) |
| Paternal job loss<br>×high income( $t-1$ ) |                   | 0.03<br>(0.21)           |                   | 0.04<br>(0.21)           |                   | 0.02<br>(0.21)            |                   | 0.16<br>(0.20)         |
| Low household<br>income( $t-1$ )           |                   | -0.07<br>(0.07)          |                   | -0.05<br>(0.07)          |                   | 0.09<br>(0.07)            |                   | 0.20***<br>(0.07)      |
| High household<br>income( $t-1$ )          |                   | -0.08<br>(0.08)          |                   | -0.05<br>(0.08)          |                   | -0.13<br>(0.08)           |                   | -0.11<br>(0.08)        |
| $Y(t-1)$                                   | 0.09***<br>(0.03) | 0.09***<br>(0.03)        | 0.15***<br>(0.03) | 0.15***<br>(0.03)        | 0.14***<br>(0.03) | 0.14***<br>(0.03)         | 0.17***<br>(0.03) | 0.17***<br>(0.03)      |
| $R^2$                                      | 0.18              | 0.18                     | 0.23              | 0.24                     | 0.11              | 0.12                      | 0.17              | 0.18                   |
| No. of observations                        | 1080              | 1080                     | 1075              | 1075                     | 1081              | 1081                      | 1081              | 1081                   |
| <b>Panel B: Mother</b>                     |                   |                          |                   |                          |                   |                           |                   |                        |
| Maternal job loss                          | 0.10<br>(0.14)    | -0.05<br>(0.25)          | -0.11<br>(0.11)   | -0.20<br>(0.16)          | -0.07<br>(0.15)   | 0.01<br>(0.31)            | -0.15<br>(0.11)   | -0.14<br>(0.18)        |
| Maternal job loss<br>×low income( $t-1$ )  |                   | 0.27<br>(0.37)           |                   | 0.23<br>(0.27)           |                   | -0.06<br>(0.36)           |                   | -0.05<br>(0.27)        |
| Maternal job loss<br>×high income( $t-1$ ) |                   | 0.20<br>(0.34)           |                   | 0.08<br>(0.24)           |                   | -0.18<br>(0.36)           |                   | 0.01<br>(0.24)         |
| Low household<br>income( $t-1$ )           |                   | -0.12*<br>(0.07)         |                   | -0.05<br>(0.07)          |                   | 0.01<br>(0.07)            |                   | 0.07<br>(0.07)         |
| High household<br>income( $t-1$ )          |                   | -0.07<br>(0.08)          |                   | -0.04<br>(0.09)          |                   | -0.07<br>(0.08)           |                   | -0.08<br>(0.07)        |
| $Y(t-1)$                                   | 0.09***<br>(0.03) | 0.09***<br>(0.03)        | 0.14***<br>(0.04) | 0.14***<br>(0.04)        | 0.16***<br>(0.03) | 0.16***<br>(0.03)         | 0.19***<br>(0.03) | 0.18***<br>(0.03)      |
| $R^2$                                      | 0.17              | 0.17                     | 0.21              | 0.21                     | 0.10              | 0.10                      | 0.20              | 0.20                   |
| No. of observations                        | 1071              | 1071                     | 1066              | 1066                     | 1072              | 1072                      | 1072              | 1072                   |

# VI. Channels



|                                | Take care of<br>children age 6<br>and less | Buy food for<br>household | Prepare and<br>cook food |
|--------------------------------|--------------------------------------------|---------------------------|--------------------------|
|                                | (1)                                        | (2)                       | (3)                      |
| <b>Panel A: Father</b>         |                                            |                           |                          |
| Paternal job loss              | 0.07<br>(0.31)                             | -0.18<br>(0.14)           | -0.08<br>(0.15)          |
| Low household income( $t-1$ )  | -0.38*<br>(0.23)                           | 0.12<br>(0.09)            | 0.30***<br>(0.10)        |
| High household income( $t-1$ ) | -0.69***<br>(0.22)                         | 0.01<br>(0.10)            | -0.17#<br>(0.11)         |
| Low household income( $t$ )    | 0.10<br>(0.20)                             | -0.01<br>(0.09)           | -0.18*<br>(0.10)         |
| High household income( $t$ )   | -0.19<br>(0.23)                            | -0.05<br>(0.10)           | -0.23**<br>(0.10)        |
| No. of observations            | 363                                        | 1300                      | 1275                     |
| <b>Panel B: Mother</b>         |                                            |                           |                          |
| Maternal job loss              | <b>0.56#</b><br>(0.36)                     | <b>0.38**</b><br>(0.17)   | <b>0.07</b><br>(0.18)    |
| Low household income( $t-1$ )  | 0.90**<br>(0.37)                           | 0.07<br>(0.10)            | 0.07<br>(0.12)           |
| High household income( $t-1$ ) | 0.01<br>(0.25)                             | 0.06<br>(0.11)            | 0.24**<br>(0.12)         |
| Low household income( $t$ )    | 0.20<br>(0.30)                             | -0.43***<br>(0.10)        | 0.17<br>(0.12)           |
| High household income( $t$ )   | -0.23<br>(0.29)                            | -0.10<br>(0.11)           | -0.01<br>(0.12)          |
| No. of observations            | 397                                        | 1332                      | 1327                     |

# VI. Channels



|                                        | Receive Any Immunizations<br>(Children under 12) |                  |                  | Receive Any Preventive Care<br>(All Children) |                   |                   |
|----------------------------------------|--------------------------------------------------|------------------|------------------|-----------------------------------------------|-------------------|-------------------|
|                                        | (1)                                              | (2)              | (3)              | (4)                                           | (5)               | (6)               |
| <b>Panel A: Father</b>                 |                                                  |                  |                  |                                               |                   |                   |
| Paternal job loss                      | -0.14<br>(0.18)                                  | -0.14<br>(0.18)  | -0.05<br>(0.28)  | -0.11<br>(0.18)                               | -0.08<br>(0.18)   | -0.03<br>(0.27)   |
| Paternal job loss×low income( $t-1$ )  |                                                  |                  | -0.45<br>(0.42)  |                                               |                   | 0.03<br>(0.44)    |
| Paternal job loss×high income( $t-1$ ) |                                                  |                  | 0.03<br>(0.44)   |                                               |                   | -0.17<br>(0.39)   |
| Low household income( $t-1$ )          | 0.11<br>(0.14)                                   | 0.13<br>(0.14)   | 0.17<br>(0.15)   | -0.28*<br>(0.15)                              | -0.33**<br>(0.14) | -0.33**<br>(0.15) |
| High household income( $t-1$ )         | 0.14<br>(0.18)                                   | 0.07<br>(0.18)   | 0.06<br>(0.20)   | -0.28*<br>(0.16)                              | -0.26#<br>(0.17)  | -0.23<br>(0.18)   |
| Low household income( $t$ )            |                                                  | -0.04<br>(0.15)  | -0.03<br>(0.15)  |                                               | 0.10<br>(0.15)    | 0.10<br>(0.15)    |
| High household income( $t$ )           |                                                  | 0.33**<br>(0.16) | 0.34**<br>(0.16) |                                               | -0.16<br>(0.15)   | -0.16<br>(0.15)   |
| No. of observations                    | 542                                              | 540              | 540              | 1104                                          | 1100              | 1100              |
| <b>Panel B: Mother</b>                 |                                                  |                  |                  |                                               |                   |                   |
| Maternal job loss                      | 0.02<br>(0.24)                                   | 0.03<br>(0.24)   | -0.41<br>(0.52)  | 0.09<br>(0.21)                                | 0.12<br>(0.21)    | 0.26<br>(0.32)    |
| Maternal job loss×low income( $t-1$ )  |                                                  |                  | 1.22*<br>(0.64)  |                                               |                   | -0.31<br>(0.61)   |
| Maternal job loss×high income( $t-1$ ) |                                                  |                  | 0.30<br>(0.63)   |                                               |                   | -0.22<br>(0.44)   |
| Low household income( $t-1$ )          | 0.09<br>(0.14)                                   | 0.11<br>(0.14)   | 0.05<br>(0.15)   | -0.33**<br>(0.15)                             | -0.37**<br>(0.14) | -0.35**<br>(0.15) |
| High household income( $t-1$ )         | 0.42**<br>(0.18)                                 | 0.35*<br>(0.19)  | 0.35*<br>(0.21)  | -0.24<br>(0.15)                               | -0.20<br>(0.16)   | -0.18<br>(0.18)   |
| Low household income( $t$ )            |                                                  | -0.02<br>(0.15)  | -0.02<br>(0.15)  |                                               | 0.09<br>(0.15)    | 0.09<br>(0.15)    |
| High household income( $t$ )           |                                                  | 0.23<br>(0.16)   | 0.23<br>(0.16)   |                                               | -0.13<br>(0.15)   | -0.13<br>(0.15)   |
| No. of observations                    | 554                                              | 552              | 552              | 1096                                          | 1094              | 1094              |

# VI. Channels



|                                | Children Age $\geq 12$ and<br>Smoke |                    | Father Smokes |                    |
|--------------------------------|-------------------------------------|--------------------|---------------|--------------------|
|                                | Coef.                               | Marginal<br>Effect | Coef.         | Marginal<br>Effect |
|                                | (1)                                 | (2)                | (3)           | (4)                |
| <b>Panel A: Father</b>         |                                     |                    |               |                    |
| Paternal job loss              | <b>1.05**</b>                       | <b>0.0013</b>      | <b>0.28**</b> | <b>0.09</b>        |
|                                | (0.45)                              |                    | (0.13)        |                    |
| Low household income( $t-1$ )  | 0.96                                |                    | 0.14          |                    |
|                                | (0.65)                              |                    | (0.09)        |                    |
| High household income( $t-1$ ) |                                     |                    | -0.07         |                    |
|                                |                                     |                    | (0.10)        |                    |
| Low household income( $t$ )    | 0.001                               |                    | -0.08         |                    |
|                                | (0.38)                              |                    | (0.09)        |                    |
| High household income( $t$ )   | 0.54                                |                    | -0.08         |                    |
|                                | (0.44)                              |                    | (0.09)        |                    |
| No. of obs.                    | 618                                 |                    | 1443          |                    |
|                                | Children Age $\geq 12$ and<br>Smoke |                    | Mother Smokes |                    |
| <b>Panel B: Mother</b>         |                                     |                    |               |                    |
| Maternal job loss              | 0.59                                |                    | 0.51          |                    |
|                                | (0.55)                              |                    | (0.42)        |                    |
| Low household income( $t-1$ )  | 0.28                                |                    | -0.34         |                    |
|                                | (0.39)                              |                    | (0.26)        |                    |
| High household income( $t-1$ ) |                                     |                    | 0.42          |                    |
|                                |                                     |                    | (0.28)        |                    |
| Low household income( $t$ )    | -0.004                              |                    | 0.15          |                    |
|                                | (0.44)                              |                    | (0.26)        |                    |
| High household income( $t$ )   | -0.10                               |                    | 0.33          |                    |
|                                | (0.33)                              |                    | (0.28)        |                    |
| No. of obs.                    | 613                                 |                    | 748           |                    |